PCAI Model Mix and Developer Sizing

DeepSeek V4.1 Flash premium tier, the accelerator tier, and what “max output tokens/second” actually measures

Prepared for the PCAI AI Specialist team | November 2026 | ModelBenchmarker results

Summary

- **The mix is now five models.** DeepSeek V4.1 Flash takes the premium agentic-coding slot (20% of output tokens), DeepSeek V4 Flash 0731 stays the high-volume coding agent (40%), Qwen3.8-27B covers chat, RAG and documents (15%), Qwen3.8-Flash-Next covers tab autocomplete and sub-agents (25%), and GLM-5.3-Flash becomes a *cross-review sidecar* that carries no traffic share.
- **V4.1 Flash is both the strongest and the fastest model in the mix.** It is the highest-ranked open-weights model on Artificial Analysis, and on our own 4x H200 engram-offload run it delivers **914 output tok/s per H200** (64 users) against GLM-5.3-Flash’s **489** (32 users, its highest straggler-clean level). GLM is kept for a different job: its hallucination rate is lower, and a second model reading the same diff catches what one model does not.
- **A Large (16x H200)** has a 23,241 tok/s decode ceiling and supports **3.03T total tokens a year**, **1.97T of it practical** at the 0.65 capacity factor — **960 to 1,600 developers** for a mixed population, 1,364 at the Airline Customer Metric. Prefill time is charged in every figure.
- **From Medium up, the question is no longer which LLM fits but how many GPUs go to LLM replicas.** We recommend devoting **one GPU, MIG-split**, to the non-LLM tier: embedding and reranker for RAG and memory applications, and a decision model for LLM task routing. That GPU pays for itself once the router downgrades roughly one request in ten.
- **We quote only straggler-clean levels.** A throughput level whose TTFT P100/P50 exceeds 2 is one slow request, not a measurement. On that rule DeepSeek V4 Flash is **1,370** output tok/s per H200, not the 1,786 its 128-user level would suggest, and V4.1 Flash is **1.9x** GLM-5.3-Flash on the same four GPUs.
- **Plan on the measured numbers**, quote the ceiling only next to the practical figure, and size to an SLO (TTFT, TPOT) rather than to a peak.

1. Updated model mix

Model	Parameters	Role	tok/s per GPU, H200 / RTX	VRAM (weights), GPUs	Mix H200/RTX	License and notes
DeepSeek V4.1 Flash 1017	552B MoE	Premium agentic coding; primary model for hard work	**914** (peak 1,090) / —	≈614 GB, **4** with engram offload (≈200B params in host RAM)	20% / 0%	MIT. Highest-ranked open-weights model on Artificial Analysis. Measured on 4x H200 with engram offload.
GLM-5.3-Flash	320B MoE, 18B active	Cross-review and second opinion; vision	489 (peak 722) / —	≈328 GB FP8, 4	sidecar (fixed 4 GPUs)	MIT. Lowest hallucination rate we run; reviews a sample of premium output rather than carrying share.
DeepSeek V4 Flash 0731	284B MoE, 13B active	Main coding agent, high volume	1,370 (peak 1,930) / 437 (peak 448)	≈167 GB FP4/FP8, 2	40% / 60%	MIT. Retired from DeepSeek’s API in favour of V4.1; weights remain open. Memory-tight on 2x RTX; DSPARK MTP cannot be enabled.
Qwen3.8-27B	27B dense	Chat, RAG context, documents, vision, fine-tune base	**1,850** (peak 3,283) / 760 NVFP4 (peak 1,424)	≈56 GB BF16, 1	15% / 15%	Apache 2.0. H200 figure is SGLang + DFlash2 + HiCache x3; RTX is vLLM NVFP4.
Qwen3.8-Flash-Next	≈120B MoE, ≈4B active (config-sized)	Tab autocomplete (FIM), sub-agents, utility	1,940 *(est.)* / 1,207 *(est.)*	≈120 GB FP8, 1 (tight)	25% / 25%	Apache 2.0. **No team benchmark**; estimates only. Confirm the model before quoting.
Custom / fine-tune	27B	LoRA on Qwen3.8-27B	≈1,000 *(est.)* / ≈700 *(est.)*	1 GPU	0%	Placeholder for a customer fine-tune.
Blended (harmonic)			**1,387** (peak 1,769)			

How to read the throughput. The first number is measured output (decode) speed on the coding task at 32K context, at the highest concurrency we ran with zero failures *and* no straggler (TTFT turn-1 P100/P50 ≤ 2). “Peak” is the same measurement at zero context, where TTFT variance is larger in relative terms but small in absolute terms. Both are per-request P50 aggregated over users and divided by GPU count. Mix % is each model’s share of *output* tokens; the blended figure is the harmonic (GPU-time weighted) mean at the stated mix. Full derivation and per-level provenance: `results/README.md` and the data pack in the working notes.

Why these models

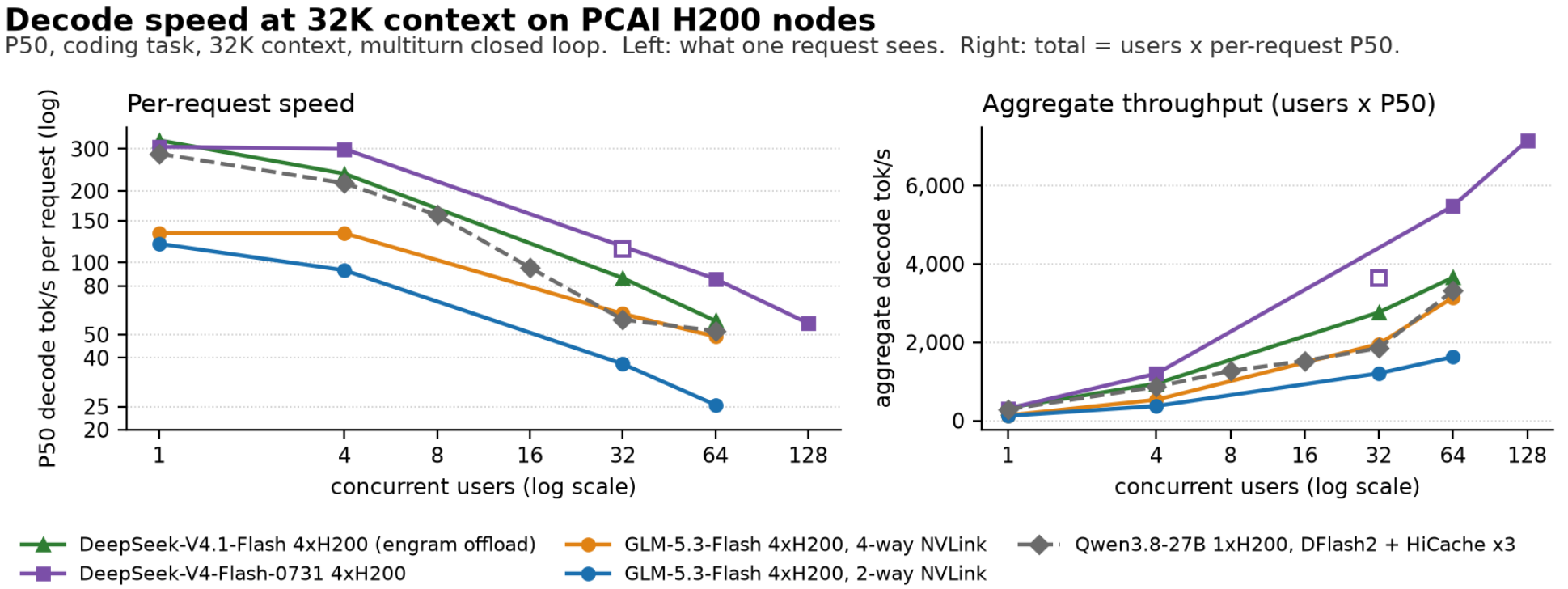
- **DeepSeek V4.1 Flash 1017** is the strongest open-weights model available to us, and it is also *cheaper per token* than GLM-5.3-Flash: 914 tok/s per H200 against 489 on the same 4-GPU footprint, a 1.9x advantage that survives the straggler test on both sides. Its ≈614 GB of weights do not fit 4x H200 (564 GB) until the engram lookup tables come out of HBM: the measured configuration keeps roughly 200B parameters of engram tables in host RAM, which is what makes four GPUs work at all — without that offload the same box has no KV pool left. An 8x H200 resident configuration should be faster and is the first benchmark to close (see §8).
- **GLM-5.3-Flash** keeps a role because raw speed is not the only axis. It has an excellent hallucination rate, serving as the cautious reviewer of DeepSeek's premium output — the two-model cross-review pattern we use internally. It does not carry a traffic share: the requirement is one 4-GPU replica, which is why it appears on Large and above and not below.
- **DeepSeek V4 Flash 0731** remains the volume workhorse: 1,370 tok/s per H200 at 32K context, 167 GB of weights so it fits two GPUs, and MIT terms even though DeepSeek retired it from their own API. On 2x RTX PRO 6000 it is memory-tight but serviceable at 437 tok/s/GPU.
- **Qwen3.8-27B** is the only dense, single-GPU model in the mix: Apache 2.0, vision-capable, 56 GB of BF16 weights, and the easiest base for a customer fine-tune. It is the cheapest way to add capacity: one GPU buys 1,850 tok/s on the SGLang + DFlash2 + HiCache x3 configuration we recommend, 1.7x what the same GPU delivers on a plain vLLM deployment (see §8).
- **Qwen3.8-Flash-Next** takes the autocomplete and sub-agent slot. We have **no team benchmark** for it. Its config sizes to roughly 120B total and 4B active parameters; the throughput figures are estimates from a comparable MoE in the same serving class — 1,207 tok/s measured on RTX, 1,940 applying our measured 1.6x H200-to-RTX ratio — and both are labelled as such. Sizing is tight either way: the BF16 checkpoint is ≈230 GB and needs two GPUs, while an FP8 build (≈116 GB) fits one H200 with only ≈7 GB of KV pool left, about 16 sequences at 32K. That is enough for autocomplete and sub-agent traffic, not for long-context chat. A 2x H200 deployment has been observed in a colleague's hosted trial; we have not measured it. **Confirm the model identity before this number leaves the building.**
- **Skipped:** Codestral (Mistral Non-Production License), MiniMax M3.1 Flash (commercial use requires notifying MiniMax and written authorization above $20M annual revenue), Gemma 4 31B (about 790 tok/s at zero context on a full H200, mixed task — roughly a quarter of Qwen3.8-27B at short prompts, and in our experience it performs worse than Qwen3.8-27B, and even Qwen 3.6-27B, on the work we care about).

2. What "maximum output tokens per second" actually measures

Every throughput number in this document is *decode* speed: tokens per second of generated text, measured from the first token to the end of the stream, with the wait for the first token excluded. Every figure here comes from a committed run under `results/`, with the derivation rule in §1 and the per-level provenance recorded beside the artifacts. That single choice explains most of the disagreement between our numbers and a vendor's headline. A request passes through three phases, and each has a different cost and a different bottleneck.

- **Prefill (input).** The prompt is processed before the first token appears. This is compute-bound and fast in aggregate: **11,500 tok/s** on DeepSeek V4.1 Flash at 32K context on 4x H200, 15,400 on GLM-5.3-Flash and 20,300 on DeepSeek V4 Flash, with the whole range 10–20k depending on sequence depth. It does not collapse at long context: GLM-5.3-Flash still holds 21,400 tok/s at 512K tokens. Prefill sets TTFT, and TTFT is what users feel.
- **Decode (output).** One token per step per sequence, so the per-request rate *falls* as concurrency rises while the aggregate *rises*. On DeepSeek V4.1 Flash a single request at 32K context gets 326 tok/s; at 64 concurrent users each gets 57 tok/s and the node produces **64 × 57 ≈ 3,650 tok/s** in aggregate. At zero context the same box does better — 68 tok/s per user, 64 × 68 ≈ 4,350 aggregate — which is exactly why the headline figure needs its context length stated. The aggregate is the number that belongs in a capacity plan; the per-request number is the one that belongs in a user-experience claim. They are not interchangeable, and quoting one as the other is the most common sizing error we see.
- **Cache (reused input).** A multi-turn conversation re-sends the same prefix on every turn. With prefix/KV caching the server skips the recompute: on DeepSeek V4.1 Flash at 32K context, TTFT falls from 2,845 ms on turn 1 to 676 ms on turn 2 (4.2x), and on GLM-5.3-Flash from 4,276 ms to 1,319 ms at four users. Cached input is nearly free on the box and *not* free on the invoice — cloud providers bill cached input at a discount rather than at zero, which is why a token budget built from list prices and a token budget built from our tok/s diverge. Cache is also where the 4x H200 boxes break at high load: between 32 and 64 users at 32K context, GLM-5.3-Flash's cached-prefix TTFT goes from 1,487 ms to 114,645 ms on the 4x H200 box while decode falls only 20–35%. The cliff is host-cache capacity, not compute: above roughly 32 users at 32K the purchasing lever is memory, not decode speed. That memory is there to be used. HiCache spills the KV pool into host RAM — 1 TB on a Small, 1.5 TB per node on Medium, 2 TB per

node on Large — and behind that into an NVMe backing store over RDMA. The effect is to turn a 141 GB HBM budget into a multi-terabyte cache tier for a few percent of the cost of another GPU, which is why the same four H200s hold 32K-context agentic sessions at all.



Source: results/deepseek_v41_flash/H200x4_engram_offload.md; results/deepseek_v4_flash_0731/H200Sx4_hicachex2.md; results/glm-5.3-flash/H200x4_hicachex4.md and H200x4_NVLink2_hicachex6.md; results/qwen_38_27b/H200_sglang_dflash2_hicachex3.md. Hollow marker: DeepSeek V4 Flash at 32 users had 2 failed requests, so that point is excluded from the line.

Figure 1. Decode speed at 32K context on the coding task, P50, multiturn closed loop. Left: what one request sees; right: aggregate = users × per-request P50, the capacity curve. DeepSeek V4 Flash (purple) is the throughput leader — 1.5x V4.1 Flash (green) per GPU — and holds its per-request rate further into concurrency, reaching 7,200 aggregate tok/s at 128 users; V4.1 Flash is the stronger model on capability and takes the premium slot. The two GLM curves are the same model on the same four GPUs with 4-way (orange) and 2-way (blue) NVLink: 1.6x apart per request at 32 users, which is what the interconnect buys when all-reduce is on the critical path. Qwen3.8-27B on one H200 (grey) tracks the four-GPU boxes to 16 users, then saturates. Hollow marker = level with failed requests, excluded from the line.

3. Maximum throughput and developers per PCAI size

Placement is chosen to maximise served throughput at the target mix, subject to whole-GPU placement and to every replica staying inside one node. Sizes follow the current PCAI Developer, Small, Medium and Large configurations; “Large expanded” is a Large plus one 16x H200 expansion rack.

PCAI size	GPU placement (GPUs)	Decode tok/s	Max total tok / yr	Practical tok / yr (0.65)	Devs: heavy / Airline / light	Est. range
Dev System, 2x RTX	V4 Flash (2)	874	0.17T	0.11T	34 / 76 / 227	50 to 90
Small, 4x RTX	V4 Flash (2), Qwen 27B (1), Flash-Next (1)	2,841	0.28T	0.18T	57 / 127 / 378	90 to 150
Small, 8x RTX	V4 Flash x3 (6), Qwen 27B (1), Flash-Next (1)	4,589	0.85T	0.55T	172 / 382 / 1,135	270 to 450
Medium, 8x H200	V4.1 Flash (4), V4 Flash (2), Qwen 27B (1), Flash-Next (1)	9,635	0.94T	0.61T	190 / 423 / 1,255	300 to 490
Large, 16x H200	V4.1 Flash (4), V4 Flash (6), Qwen 27B (3), Flash-Next (3)	23,241	3.03T	1.97T	614 / 1,364 / 4,052	960 to 1,600
Large expanded, 32x H200	V4.1 Flash x2 (8), V4 Flash (12), Qwen 27B (6), Flash-Next (6)	46,482	6.05T	3.93T	1,228 / 2,729 / 8,104	1,900 to 3,200
Large, 16x RTX	V4 Flash x6 (12), Qwen 27B (2), Flash-Next (2)	9,178	1.69T	1.10T	344 / 764 / 2,270	540 to 890

Decode tok/s is the measured output rate at the target mix, with no input charged — it is the input to every other column. **Max total** charges prefill: for each model, total $= 1/$ ($0.15/\mathrm{D} + 0.1275/\mathrm{P}$) at its measured aggregate decode and prefill rates, combined at the target mix and multiplied by 31.536M seconds a year. The 0.15 and 0.1275 are the output and uncached-input shares of traffic at an 85/15 split and an 85% cache-hit rate; see the equation below. **Practical** applies the 0.65 useful-capacity factor. **Devs by profile** are sized to the peak hour from the same total rate: output (0.15 × total) divided by each profile"s peak demand. **Est. range** mixes profiles: low end 40% heavy agentic / 50% Airline Customer Metric / 10% light, high end 10% / 50% / 40%.

What binds each row. Dev, both Small rows and Large RTX are limited by DeepSeek V4 Flash on the RTX pair; Medium by the same model on two H200s; Large and Large expanded by the four-GPU V4.1 Flash replica. The binding model is the one to widen first.

Tokens per day, once prefill is charged

The sizing table turns output tok/s into total tokens by dividing by 0.15, which quietly assumes input is free. It is not: on a saturated box prefill and decode draw on the same time budget, and a prefill step is a decode step the box does not take. Charge it explicitly. Per token of total traffic, a fraction f_o is output and costs f_o/D seconds at the aggregate decode rate D; a fraction $f_i(1-h)$ is uncached input and costs that much at the aggregate prefill rate P; cached input is free. So

$$\text{total tok/s} = 1 \;/\; (\; 0.15/\mathrm{D} + 0.85 \times (1 - 0.85) \;/\; \mathrm{P}\;) = 1 \;/\; (\; 0.15/\mathrm{D} + 0.1275/\mathrm{P}\;)$$

with D the measured aggregate decode rate, P the measured aggregate prefill rate, and 0.15 / 0.85 the output / input split. Setting the cache-read cost to zero is deliberate: a cached token costs a KV read — 900 B at 4.8 TB/s is about 0.2 µs — against 87 µs of prefill, so it moves the answer by under half a percent. P is the single-user aggregate prefill, the conservative choice, because the measured aggregate rises slightly with concurrency (11,500 to 11,900 tok/s on V4.1 Flash).

PCAI size	Total tok/s	Tokens / day	Tokens / month	Tok / day at 0.65	Tok / month at 0.65
Dev System, 2x RTX	5,374	0.46B	13.9B	0.30B	9.1B
Small, 4x RTX	8,957	0.77B	23.2B	0.50B	15.1B
Small, 8x RTX	26,870	2.32B	69.6B	1.51B	45.3B
Medium, 8x H200	29,712	2.57B	77.0B	1.67B	50.1B
Large, 16x H200	95,928	8.29B	248.6B	5.39B	161.6B
Large expanded, 32x H200	191,856	16.58B	497.3B	10.77B	323.2B
Large, 16x RTX	53,740	4.64B	139.3B	3.02B	90.5B

Rates are the measured per-replica decode and prefill figures from the model table; Qwen3.8-Flash-Next has no prefill measurement, so DeepSeek V4 Flash's prefill stands in as a proxy. Tokens per day is total tok/s × 86,400, per month × 2.592M (30 days).

Two things fall out. First, the prefill tax runs from 5% to 21% and it is largest where decode is fastest: V4.1 Flash gives up 21% of its decode budget to input, DeepSeek V4 Flash 19%, Qwen3.8-27B 13%, GLM-5.3-Flash 10%. A fast decoder spends proportionally more of its time on input, so the ordering by total tokens is not the ordering by decode tok/s. Second, the two tables agree by construction: 5.39B tokens a day is 1.97T a year, which is the Practical total for a Large.

VRAM budget, and what it buys in concurrency

Replica	GPUs	Weights / GPU	KV B / token	KV pool / GPU	Tokens in VRAM	Sequences at 32K
DeepSeek V4.1 Flash, engram in host RAM	4	67 GB	900 B	30 GB	35.3M	1,077
GLM-5.3-Flash	4	73 GB	7.0 kB	45 GB	6.9M	210
DeepSeek V4 Flash 0731	4	68 GB	6.8 kB	43 GB	6.8M	207
Qwen3.8-27B	1	23 GB	32 kB	99 GB	3.3M	99
Qwen3.8-Flash-Next (FP8)	1	116 GB	13.8 kB	7 GB	0.5M	16

Estimated with the repository's memory model at FP8 KV and each entry's catalog `mem-fraction-static`. Tokens in VRAM is the KV pool divided by bytes per token, summed over the replica; sequences at 32K divides that by 32,768. Two things fall out. V4.1 Flash's KV is about eight times cheaper per token than GLM-5.3-Flash's (900 B against 7 kB), so the same four GPUs carry five times the concurrent context — the premium model is also the cheaper one to run hot. And Qwen3.8-Flash-Next on one H200 has almost no KV headroom, which is why it belongs on short-context traffic only. Weights are the measured checkpoint sizes; the engram row is what the host-table offload buys (without it the same box shows no KV pool).

4. Beyond LLM tokens: the accelerator tier

At Dev and Small sizes the question is which LLM fits on the GPUs you have. From Small upward it inverts: how many GPUs go to LLM replicas, and how many to the rest of the pipeline — routing, retrieval, voice. The exchange rate is concrete. One DeepSeek V4.1 Flash replica is 4x H200 and produces 3,654 tok/s at 64 users; one GLM-5.3-Flash replica is 4x H200 and produces 1,955. The same four GPUs buy the entire support tier below, with a GPU left over.

Model (catalog name)	Role	Parameters / precision	GPU request	Memory	What it buys
`vllm-sr/d3-turbo`	Decision / routing: premium vs cheap tier, per request	26.1B, bf16, fp32 readout	1 (planned)	≈52 GB weights	Premium decode avoided on every request routed down; structured readout (choice, score, box, track), not prose, so its own decode cost is negligible
`Qwen3-VL-Embedding-8B`	RAG embeddings, text/image/video	8B, bf16 or FP8	co-located @ 0.30	16 Gi	Retrieval quality on residual HBM; 4,096-dim, 8,192-token chunks
`Qwen3-VL-Reranker-8B`	RAG reranking	8B, bf16	co-located @ 0.275	16 Gi	Fewer, better chunks per LLM call — the cheapest way to cut input tokens
`cohere-transcribe-03-2026`	ASR (speech in)	2B	co-located @ 0.15	20 Gi	Voice input without a dedicated card
`fishaudio/s2-pro`	TTS (speech out)	4B-class	co-located @ 0.30	32 Gi	Voice output on residual HBM
`whisper-large-v3-turbo`	ASR alternative	—	1 dedicated only	20 Gi	Fallback ASR; no shared variant in the catalog yet

Source: `src/model_benchmarker/webapp/seed_catalog.json`. "Co-located @ x" is a serving entry with `resource_request_gpu: 0` and `--gpu-memory-utilization x`: it shares an existing GPU instead of taking one. No auxiliary model here has a team throughput benchmark; these are catalog serving configurations, not measurements.

Co-location is the headline. The shared variants bound themselves with `--gpu-memory-utilization`: embedding 0.30 + reranker 0.275 + ASR 0.15 = 0.725 of one H200's 141 GB, so all three fit inside a single GPU's memory budget; adding TTS (0.30) takes the stack to 1.025, i.e. two GPUs' residual memory. The slices match what the LLM entries leave unclaimed — `mem-fraction-static` on H200 runs from 0.70 (DeepSeek V4.1 engram offload) to 0.90 (the DeepSeek V4 Flash HiCache entry), leaving 14–42 GB per GPU — a 0.10–0.30 slice. The full 0.30 embedding slice therefore co-locates only on the 0.70–0.85 nodes, which is where the mix models sit. In other words, a PCAI unit already paid for can carry a full RAG and voice stack without giving up an LLM replica.

The routing case is where the tokens are. In the mix above, the two premium classes carry 60% of output tokens. On a Large that is 0.60 × 16,445 ≈ 9,867 tok/s of premium decode. A decision model that downgrades a fraction D of premium-bound requests frees roughly D × 9,867 tok/s (an upper bound: it assumes downgraded requests have premium-average output length). At D = 20% that is 1,973 tok/s — about one GLM-5.3-Flash replica, or half of a V4.1 Flash replica. The tier costs one GPU — the router itself, because the embedding, reranker and voice slices ride on residual HBM rather than on a card of their own — so it breaks even at a **9% downgrade rate** if the freed capacity would otherwise be V4.1-class, or 5% if it is GLM-class. Below that it is a quality play; above it, it is free capacity.

A strong RAG pipeline changes the sizing question because it buys answer quality with residual HBM instead of parameters: embedding (4,096-dim, 8,192-token chunks, 512 overlap) plus reranker is 0.575 of one co-located GPU, and better retrieval means fewer and better chunks per call, which is a direct cut to the 85% of tokens that are input. The repository's RAG transcripts (`results/RAG/`) benchmark the retrieval service alone — about 51 requests/s at 100 concurrent users, top-k 10, reranker off — in requests per second, a different unit from the chat tok/s here, and the two are not comparable.

What these numbers are not. `results/` covers the mix LLMs and the RAG service; no team benchmark exists for any auxiliary model, and the GPU/memory columns are catalog requests. The decision model has no seed-catalog serving entry: its ≈52 GB of bf16 weights fit a 141 GB H200 with room to spare, but plan a whole GPU for KV and fp32-readout headroom (262,144-token max input, 32,768 validated, 128 max sequences), and remember that a router reads every prompt it triages. Co-location costs TTFT under contention. Pilot before quoting.

5. The biggest swing factor: peak hour versus monthly volume

At the 0.65 capacity factor a Large H200 supports about 164B total tokens a month. Spread evenly over the month that is roughly 5,500 developers at the Airline Customer Metric. For calibration, that engagement's 92B total tokens a month sits between a Medium (51B a month at the same practical factor) and a Large (164B) — a Medium-plus estate, and a useful sanity check that the sizing lands where the observed estate does. Sized to the peak hour it is 1,364 — and that figure carries no capacity factor, so on the same 0.65 basis it is 887. The 6x between 5,500 and 887 is the real gap, and it exists because developers use only about 24% of the month's hours while the peak hour runs 1.5x the average. If a customer spans time zones, or runs CI agents and review bots overnight, the real number moves toward the volume figure — and that off-hours capacity is a natural pitch for selling tokens in buckets rather than seats.

6. Guidance for the PCAI Business Case workbook

1. **Blend with a harmonic mean.** If the Model Mix tab computes SUMPRODUCT(mix, tok/s), it overstates throughput. Use `1/SUMPRODUCT(mix/tok/s)`. On this mix SUMPRODUCT gives 1,493 against a correct 1,387 — 8% too high — and the error grows with the spread of rates in the mix.
2. **Apply the straggler test before you quote a rate.** A level whose TTFT P100/P50 exceeds 2 is one slow request, not a measurement. This is what moved DeepSeek V4 Flash from 1,786 to 1,370 tok/s/GPU, and it is the difference between a defensible number and a number a customer can break.
3. **Keep units consistent.** The tok/s column measures output only, while consumption benchmarks like the Airline Customer Metric are total tokens. Comparing them directly understates developers supported by about 6.7x. Convert demand to output tokens and charge uncached input against the measured prefill rates.
4. **Place models, do not blend them.** At small sizes the mix comes in whole GPUs: a Small cannot serve a 40% share of a 2-GPU model that is already the bottleneck. Model placement per size, not a single blended number, decides what a box actually serves.

5. **Expect the blended rate to sit near 1,390 per H200.** Short-prompt, maximum-batch figures for much smaller models land in the 4,600 range and are not comparable to these. For a theoretical headline use the peak blend of 1,769, but plan on the measured numbers.
6. **Fix the Llama 3.3 70B VRAM if the old row stays anywhere**: 140 GB is the BF16 size; FP8 is about 70 GB.

7. Points for regulated buyers

- **All five models come from Chinese labs** (Z.ai, DeepSeek, Alibaba), even though the licenses are permissive. For Fed, and for some financial-services and healthcare accounts, the US-origin swap is NVIDIA Nemotron 3.5: Super VL (120B, 12B active) in the premium slot — its NVFP4 build fits one H200 and peaked at 1,860 output tok/s on a single Blackwell GPU — and Lightning (30B, 3B active) for utility work, at 4,837 output tok/s maximum on a B200. Super VL scores 26 on the open-weights intelligence index we track against 41.8 for GLM-5.3-Flash on the same October 2026 version, and we have not benchmarked either Nemotron model. Test before quoting numbers.
- **The regulated swap costs more than a licence change.** Dropping V4.1 Flash gives up the 914 tok/s/GPU premium tier and the cross-review pair, and the Nemotron replacement has not been measured on H200 by us. Size the swap as a separate exercise, not as a footnote.
- **Retirement risk is a procurement point.** DeepSeek retired V4 Flash 0731 from its own API; the weights are MIT and we hold them, but a customer building on a hosted endpoint should know the model may not stay there.

8. Recommendations beyond the corrections

1. **Quote an SLO, not a peak.** The harness can report goodput — the share of successful requests meeting a joint TTFT/TPOT target (e.g. TTFT ≤ 2 s, TPOT ≤ 50 ms). A box that serves 16,445 tok/s at a 40-second TTFT serves nobody. Every sizing claim in this document should eventually carry the SLO it was measured at.
2. **Close the two benchmark gaps before this circulates widely.** (a) V4.1 Flash resident on 8x H200 without engram offload — expected to beat the 914 tok/s/GPU we get with offload, and it is the configuration a customer will actually buy; (b) Qwen3.8-Flash-Next, whose 25% share rests on estimates rather than on a measurement of that model.
3. **Check the configuration before the model.** Qwen3.8-27B on one H200 serves 1,088 tok/s under vLLM and 1,850 under SGLang with DFlash2 and HiCache x3 — a 70% gain from the serving stack alone. We quote the conservative number in the table; the point for a customer is that the software stack is worth as much as a model upgrade.
4. **Add a cache-hit sensitivity band.** The 85% cache-hit assumption drives the 85/15 input/output split, which drives every total-token figure. A customer at 60% cache hit sees materially less served throughput. Show the band, not a point.
5. **Do not compare across concurrency levels.** Comparing DeepSeek at 128 users with GLM at 32 overstates the gap by 30% and flatters whichever model was measured at the higher load. Pin the level, publish the level, and put it in the footnote.
6. **Above about 32 users at 32K, buy cache, not decode.** The 4x H200 boxes lose cached-prefix TTFT by one to two orders of magnitude between 32 and 64 users (1,487 to 114,645 ms) while decode falls only 20–35%. For a 32K-context agentic population, host-cache capacity is a sizing line item, not a tuning detail.
7. **A five-model mix fits 16x H200 only at reduced service.** Minimum replicas are 4 + 4 + 2 + 1 + 1 = 12 GPUs, so it fits — but at about 74,300 total tok/s instead of 95,900, a 23% cut (§3). Honouring the full premium, reviewer and volume shares together is the 32-GPU conversation.

Appendix A — Assumptions

Input	Value	Source
Throughput metric	Output (decode) tok/s, coding task, 32K context, highest concurrency with zero failures and TTFT P100/P50 ≤ 2	Team model-benchmarker results
Peak column	Same, at zero context (TTFT variance tolerated)	Team model-benchmarker results
Input / output split	85 / 15	Customer discovery data (Cursor users, Sep 2026)
Cache hit on input	85%	Assumption, pending sensitivity band
Prefill (input) rate	Team-measured aggregate prefill, reduced 25% for the nominal-context caveat	`results/README.md` derived table
Prefill charge	Explicit: total $= 1/$ ($0.15/\mathrm{D} + 0.1275/\mathrm{P}$) at the measured decode and prefill rates	§3 of this document
Heavy agentic profile	66M total tokens / dev / month (15 tasks a day × 200K tokens × 22 days)	SitePoint 2026 AI coding cost model, upper end
Airline Customer Metric	29.7M total tokens / dev / month (3,100 devs, 92.16B tokens / month)	Airline customer PCAI engagement, observed
Light assist profile	10M total tokens / dev / month (completions plus chat)	Assumption
Working pattern	22 days × 8 hours, peak hour 1.5x average	Assumption
Capacity factor	0.65	PCAI Large reference economics
Host cache (HiCache)	1 TB on a Small; 1.5 TB per node on Medium; 2 TB per node on Large; NVMe backing store over RDMA	Deployment configuration
DeepSeek V4 Flash on 2x H200	80% of the measured 4x H200 per-GPU rate	Estimate, not measured
Qwen3.8-Flash-Next on H200 / RTX	1,940 / 1,207 tok/s/GPU, scaled from a comparable MoE in the same serving class	Estimate, not measured — confirm the model first
Decision model sizing	One whole H200 for a 26.1B bf16 router	Model card in `utils/pcai_models.py`; no catalog serving entry, not benchmarked

Appendix B — Sources

- **Team benchmarks:** `results/deepseek_v41_flash/H200x4_engram_offload.md`; `results/glm-5.3-flash/` (H200x4_hicachex4, H200x4_NVLink2_hicachex6); `results/deepseek_v4_flash_0731/` (H200Sx4_hicachex2, RTXPRO6000x2_hicachex16); `results/qwen_38_27b/` (H200_sglang_dflash2_hicachex3, RTXPRO6000_vllm_nvfp4); derived prefill table in `results/README.md`.
- **Repository:** github.com/ai-solution-eng/tools/tree/main/model-benchmarker/results; serving configurations in `src/model_benchmarker/webapp/seed_catalog.json`.
- HPE Private Cloud AI QuickSpecs and family configurations: hpe.com/us/en/collaterals/collateral.a50009216enw.html
- GLM-5.3-Flash size and layout: glm5.app/blog/glm-5-3-flash-parameters; huggingface.co/zai-org/GLM-5.3-Flash. Intelligence index (Oct 2026): theopenweights.com/leaderboards/llm-intelligence.
- DeepSeek V4 Flash weights and hardware: yottalabs.ai/post/deepseek-v4-flash-hardware-requirements-gpu-memory-2026; API retirement: yottalabs.ai/post/glm-5-3-flash-vs-deepseek-v4-flash-2026.
- DeepSeek V4.1 Flash memory and ranking: yottalabs.ai/post/deepseek-v4-1-flash-hardware-requirements-gpu-memory-2026; artificialanalysis.ai/models.
- Qwen3.8-27B weight size: datalearner.com/en/benchmark-compare/qwen3-8-flash-next/qwen3-8-27b. Qwen3.8-Flash-Next: model config priced in `src/model_benchmarker/memory_model/configs.py`; no public benchmark adopted.
- Qwen3.8-Flash-Next: model config in `src/model_benchmarker/webapp/hf_configs/`; no public benchmark adopted.
- Codestral license: learn.engineering.vips.edu/ai-models/mistral-codestral. MiniMax M3.1 Flash license: pasqualepillitteri.it/en/news/26561/minimax-m3-1-flash-weights-hugging-face-license.
- Nemotron 3.5: artificialanalysis.ai/articles/nemotron-3-5-super-and-bolt; lambda.ai/inference-models/nvidia/NVIDIA-Nemotron-3.5-Super-VL-120B-A12B; lambda.ai/inference-models/nvidia/nvidia-nemotron-3.5-lightning-30b-a3b-nvfp4; orcarouter.ai/blog/nemotron-3-5-super-vl-deployment-cost.
- Heavy agentic usage profile: sitepoint.com/ai-coding-tools-cost-analysis-roi-calculator-2026.